LECTURE NOTE.

CODE: MAT 102

TITLE: Elementary Math II (Elementary Calculus)

OUTLINE.

A. FUNCTIONS: - Concept and definition, Examples — polynomials, exponential, logarithmic and trigonometric fns. Graphs and their properties.

B. DIFFERENTIAL ~~EQUATION~~ CALCULUS:
- Differentiation by 1st principle
- Techniques of differentiation.
- Sum, product, quotient and chain rules.
- Implicit differentiation
- Standard Derivatives.
- Higher Order derivatives
- L'Hopitals Rule
- Applications of Differentiation:

C. INTEGRATION:
- As anti-derivative
- The fundamental theorem of integral Calculus
- Applications of Integration

D. PLANE ANALYTIC GEOMETRY and CONIC SECTIONS.

(A) FUNCTIONS:

Defn: Let X and Y be two sets, a fn from X to Y is a rule that assigns to every element of X a unique element of Y. If f is a fn from A to B, we write it as:

$$f: A \longrightarrow B \quad \text{or} \quad A \xrightarrow{f} B$$

NB (Remarks)

(a) The sets A and B in the instance above are called Domain and Codomain respectively of the fn.

(b) If $x \in A$ is assigned to $y \in B$, then we call y the image of x under the fn, f. Then we denote that as;

$$y = f(x)$$

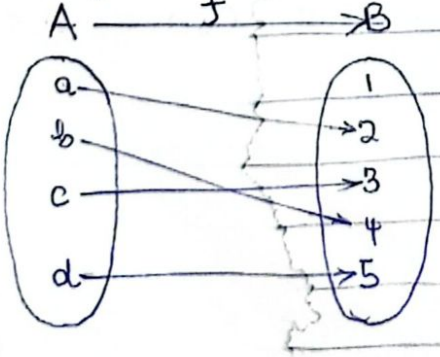
$\Rightarrow$ f takes x to y.

* Note that x is called the pre-image of y.

(c) We call y the dependent variable and x the independent variable.

(d) The subset of B that contains all the images of some elements of A under f is called the RANGE (or IMAGE) of the fn, f. We denote that by imf.

Diagrammatically;

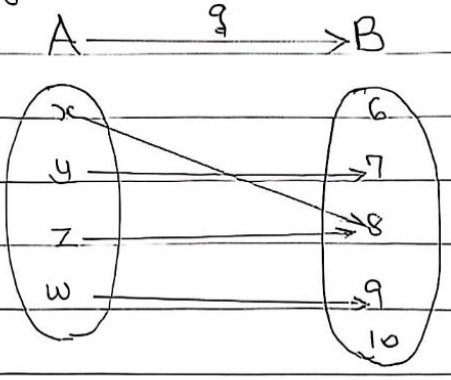


$f(a) = 2$
$f(b) = 4$
$f(c) = 3$
$f(d) = 5$

$\rightarrow$ f is a fn since every element of A has image and

no element is mapped to more than one images.

Diagram (2)

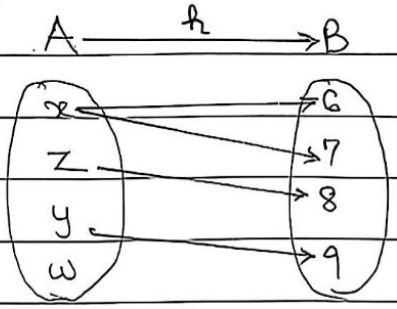


g is also a fn, and

$g(x) = 8$
$g(y) = 7$
$g(z) = 8$
$g(w) = 9$

⇒ Two elements of A can have the same image but an element of A cannot have two images.

Diagram (3):



$h: A \to B$ is not a fn bc $h(x) = 6$ and $h(x) = 7$.

⇒ x has two images, its image is not unique.

Note:

From example diagrams (1) and (2):

The range of (1) = {2, 3, 4, 5}

The range of g in (2) = {7, 8, 9}.

REAL-VALUED FUNCTION

This is a fn whose co-domain is the set of real numbers ($\mathbb{R}$). That is; $f: A \to \mathbb{R}$ is called a real-valued fn. We will concentrate on fns whose domain and co-domain are set of $\mathbb{R}$ or its subset.

We can represent ~~fns~~ by formulars. For instance:

① $f(x) = 2x$; f takes a real no to its double.

② $f(x) = x^2 + 1$

③ $f(x) = 5x^2 - 6x + 2$ etc.

EVALUATION ~~TYPES~~ OF FUNCTIONS

To evaluate $f(x)$ at the point $x = a$ implies to find the value of $f(a)$.

Examples:

Given that $f(x) = x^2 + 1$, evaluate

(a) $f(0)$ (b) $f(2)$ (c) $f(-\frac{1}{2})$ (d) $f(t)$ (e) $f(x+1)$

(f) $f(x+h)$.

Soln:

(a) $f(0) = 0^2 + 1 = \underline{1}$

(b) $f(2) = 2^2 + 1 = \underline{5}$

(c) $f(-\frac{1}{2}) = (-\frac{1}{2})^2 + 1 = \frac{1}{4} + 1 = \underline{1\frac{1}{4}}$

(d) $f(t) = t^2 + 1$

(e) $f(x+1) = (x+1)^2 + 1$

$= x^2 + 2x + 1 + 1$

$= \underline{x^2 + 2x + 2}$

(f) $f(x+h) = (x+h)^2 + 1$

$= \underline{x^2 + 2xh + h^2 + 1}$

TYPES OF FUNCTIONS

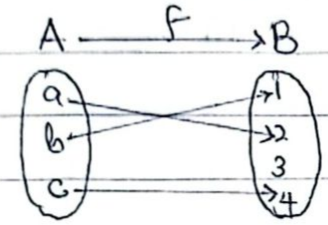
(A) One-to-One Fns (Injective).

A fn, $f: A \to B$ is called injective if element of A is mapped to only one element of B. That is, every image has only one pre-image.

→ if $f(x) = f(y)$ then $x = y$.

E.g: $f(x) = 2x$, $f(x) = x$ etc.

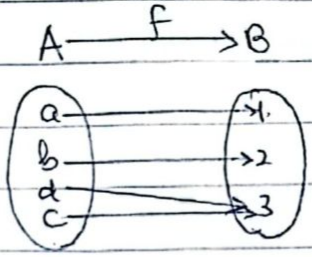
Diagram:



(B). Onto Functions (Surjective)

A fn $f: A \to B$ is onto if every element of B has a pre-image in A. That is, the range is equal to the co-domain.

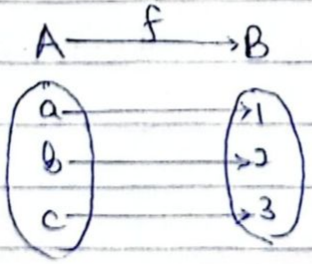
Diagram:



(C) BIJECTIVE FUNCTIONS

A fn that is both injective and surjective is called a bijective fn.

Diagram:



(D) EVEN & ODD FUNCTIONS

A fn, $f(x)$ is said to be even if

$$\boxed{f(-x) = f(x)}$$ for all x in the domain.

and is called odd if

$$\boxed{f(-x) = -f(x)}$$

Examples:

* x^2, $\cos x$, $|x|$ etc. are even fns
* x^3, $\sin x$ etc are odd fn.

Ex①: Show that:

(a) $f(x) = x^2$ is even

(b) $f(x) = x^3$ is odd.

Soln:

(a) $f(x) = x^2$

but

$$f(-x) = (-x)^2 = x^2 = f(x)$$

since $f(-x) = f(x) \Rightarrow f(x) = x^2$ is even ∎

(b) $f(x) = x^3$

$\rightarrow f(-x) = (-x)^3$

$= -x \times -x \times -x$

$= -x^3$

$= -f(x)$

$\therefore f(x) = x^3$ is odd ∎

Exercise 1: What type of fn is $f(x) = 2x^5 + x$; Even/odd?

Soln:

(E) Piecewise Fns

A piecewise fn is a fn that is defined on a sequence of intervals of inputs. ~~Eg~~

Examples:

(1) $f(x) = \begin{cases} x^2, & x < 1 \\ \sqrt{x}, & x \geq 1 \end{cases}$

(2) $f(x) = \begin{cases} 4x, & x < 2 \\ 3x - 1, & x \geq 2 \end{cases}$

(3) $f(x) = \begin{cases} 2x - 5, & x \leq -1 \\ x + 8, & -1 < x \leq 5 \\ -6x, & x > 5 \end{cases}$

Problems:

Let the fn $f: \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 4x - 7 & \text{if } x \geq 3 \\ x^2 - 3 & \text{if } -3 \leq x \leq 3 \\ 2x + 5 & \text{if } x < -4. \end{cases}$$

Evaluate;

(a) $f(-3)$ (b) $f(-1)$ (c) $f(4)$ (d) $f(-9/2)$.

Solⁿ:

(a) $f(-3) = ?$

For $x = -3$; $f(x) = x^2 - 3$

$\Rightarrow f(-3) = (-3)^2 - 3 = 6$ //

(b) $f(-1) = ?$; $f(x) = x^2 - 3$

$\Rightarrow f(-1) = (-1)^2 - 3 = -2$.

(c) $f(-9/2) = ?$; $f(x) = 2x + 5$

$\Rightarrow f(-9/2) = 2(-9/2) + 5 = -4$

COMPOSITION OF FUNCTIONS.

Given two fns $f: A \longrightarrow B$ and $g: B \longrightarrow C$, the fn $h: A \longrightarrow C$ given as $h(x) = (g \circ f)(x)$ is called the COMPOSITION of f and g from A to C.

where;

$$(g \circ f)(x) = g \circ f = g(f(x))$$

and it can go on to 3 compositions or more.

$$(g \circ f \circ h)(x) = g[f(h(x))].$$

Examples:

Define $f, g, h: \mathbb{R} \longrightarrow \mathbb{R}$ by $f(x) = |x|$, $g(x) = x+2$ and $h(x) = \cos x$. Compute the following.

(a) $h \circ f$ (b) $f \circ h$ (c) $f \circ f$ (d) $h \circ g$ (e) $g \circ g$ (f) $f \circ g$ (g) $g \circ f$.

Soln:

(a) $h \circ f = h(f(x))$
where $h(x) = \cos x$
$\Rightarrow h(f(x)) = \cos(f(x))$
$= \cos(|x|)$

(c) $f \circ f = f(f(x))$
where $f(x) = |x|$
$\Rightarrow f(f(x)) = |f(x)|$
$= ||x|| = |x|$

(b) $f \circ h = f(h(x))$
where $f(x) = |x|$
$\Rightarrow f(h(x)) = |h(x)|$
$= |\cos x|$

(d) $h \circ g = h(g(x))$
but $h(x) = \cos x$
$\Rightarrow h(g(x)) = \cos(g(x))$
$= \cos(x+2)$

Exercise 2: Do the rest.

INVERSE OF FUNCTIONS.

Defn: Let $f: A \longrightarrow B$ be a fn, then the inverse of f written as f^{-1} is given as $f^{-1}: B \longrightarrow A$ if f^{-1} is a fn itself. That is, if f is invertible.

Examples

① Let $A = \{a, b, c, d\}$ and $b = \{1, 2, 3, 4, \}$. Suppose the relation R_1 and R_2 are fns from A to B,

$$R_1 = \{(a,1), (b,1), (c,2), (d,4)\}$$

$$R_2 = \{(a,2), (b,1), (c,3), (d,4)\}$$

find: (a) R_1^{-1} (b) R_2^{-1}. (c) Determine whether R_1^{-1} & R_2^{-1} are fns.

Soln:

(a) $R_1^{-1} = \{(1,a), (1,b), (2,c), (4,d)\}$

(b) $R_2^{-1} = \{(2,a), (1,b), (3,c), (4,d)\}$

(c) * R_1^{-1} is not a fn because R_1 is not 1-1 and as a result the element "1" in R_1^{-1} has two images.

* R_2^{-1} is a fn.

N/B: A fn, $f(x)$ is invertible if and only if it is bijective.

② Find the formula for the inverse of the following fns

(a) $f(x) = \sqrt{2+5x}$ (b) $f(x) = 3x + 2$.

Soln:

(a) $f(x) = \sqrt{2+5x}$, Set $f(x) = y$

$\Rightarrow y = \sqrt{2+5x}$

make x the subject; $x = \dfrac{y^2 - 2}{5}$

interchange x and y

$\Rightarrow y = \frac{x^2-2}{5}$

$\therefore f^{-1}(x) = \frac{x^2-2}{5}$

(b) $f(x) = 3x+2$

$\Rightarrow y = 3x+2$

$\Rightarrow x = \frac{y-2}{3}$

$\therefore f^{-1}(x) = \frac{x-2}{3}$ //

Domain of Functions

Defn: The Domain of a fn, $f(x)$ is the set of values of x for which the fn, $f(x)$ is defined or real.

Examples

Find the domain of the following fns.

(1) $f(x) = \frac{2x+1}{3x}$ (2) $g(x) = \sqrt{1-x}$ (3) $h(x) = \frac{x+2}{x^2-1}$

(4) $f(x) = x^2+2x-1$ (5) $f(x) = \log(x+1)$ (6) $f(x) = \sin x$

(7) $f(x) = 1/\sin x$.

Soln:

(1) For rational fns, ensure the denominator is not zero.

$\rightarrow$ Dom $(f) = \{x \mid 3x \neq 0\}$, then solve the eqn.

$= \{x \mid x \neq 0\}$ or $\{\mathbb{R} \setminus 0\}$ or $(-\infty, 0) \cup (0, +\infty)$.

(2) For radical fn, ensure the value under the root is not negative, i.e, it must be ≥ 0.

Hence, for $g(x) = \sqrt{1-x}$

$\Rightarrow$ Dom $(g) = \{x \mid 1-x \geq 0\}$

$= \{x \mid x \leq 1\}$ or $(-\infty, 1]$

(3) $h(x) = \frac{x+2}{x^2-1}$; Dom $(h) = \{x \mid x^2-1 \neq 0\}$

$= \{x \mid (x+1)(x-1) \neq 0\}$

$= \{x \mid x \neq 1$ and $x \neq -1\}$

$= (-\infty, -1) \cup (-1, 1) \cup (1, +\infty)$

(4) For a straight polynomial or algebraic polynomials, the domain is the whole $\mathbb{R}$. Hence, $f(x) = x^2 + 2x - 1$

$\Rightarrow$ Dom $(f) = \mathbb{R}$

(5) $f(x) = \log(x+1)$

For a log fn, the value under the must ~~not~~ be +ve.

$\Rightarrow$ Dom $(f) = \{x \mid x+1 \geq 0\}$

$= \{x \mid x > -1\}$ or $(-1, +\infty)$

(6) $f(x) = \sin x$; For sine fn, it is defined everywhere

$\Rightarrow$ Dom $(f) = \mathbb{R}$.

(7) $f(x) = \dfrac{1}{\sin x}$

Dom $(f) = \{x \mid \sin x \neq 0\}$

$= \{x \mid x \neq 0, 180, 360, \ldots\}$

$= \{x \mid x \neq 180n, n \in \mathbb{Z}\}$.

Range of a function:

The range also called the image is the subset of the codomain consisting of every element that is the image of some elements in the domain. That is the possible values of $f(x)$ for the values of x in the domain.

Examples

Find the range of the fns: ① $f(x) = \sqrt{1-x}$

② $g(x) = \dfrac{2x+1}{3x}$.

Soln:

① Since $f(x) = \sqrt{1-x}$ is only defined for $1-x \geq 0$.

$\Rightarrow$ the Range ≥ 0

∴ Range of $f(x) = [0, +\infty)$.

② The range of $g(x) = \frac{2x+1}{3x}$ is the whole $\mathbb{R}$ since we can obtain any value of $\mathbb{R}$ depending on the value of x from the domain.

LIMITS OF A REAL-VALUED FUNCTION:

Defn: Given a real-valued fn, $f(x)$, the limit of f as defined as x tends to a point say 'a'. We denote that as $\lim_{x \to a} f(x) = L$ i.e "the limit of $f(x)$ as $x \to a$ is L".

Mathematically, for a given fn, $f(x)$ the limit $\boxed{\lim_{x \to a} f(x) = f(a)}$ if a is in the domain of $f(x)$.

N/B: If a is not in the domain (i.e the direction of substitution of $x = a$ is undefined.) then we will try simplification of the function or the L'Hopital's rule.

Examples

Evaluate the following limits:

① $\lim_{x \to -1} \frac{x^2-1}{x+1}$ ② $\lim_{x \to 3} (10x^2 + x - 8)$ ③ $\lim_{x \to 2} \frac{4-x^2}{2-x}$

④ $\lim_{x \to 1} \frac{x^2-x+2}{x^3+x^2-3}$ ⑤ $\lim_{x \to 0} \frac{\sqrt{x^2+1}-1}{x^2}$

Soln:

② $\lim_{x \to 3} (10x^2 + x - 8) = 10(3)^2 + 3 - 8 = 85$

① $\lim_{x \to -1} \frac{x^2-1}{x+1} = \lim_{x \to -1} \frac{(x+1)(x-1)}{x+1} = \lim_{x \to -1} \frac{x-1}{1} = -1-1 = -2$

③ $\lim_{x \to 2} \frac{4-x^2}{2-x} = \lim_{x \to 2} \frac{2^2-x^2}{2-x} = \lim_{x \to 2} \frac{(2+x)(2-x)}{2-x} = \lim_{x \to 2} 2+x = 4$

⑤ $\lim_{x \to 0} \frac{\sqrt{x^2+1} - 1}{x^2}$

By direct substitution, we have $0/0$. We can try to rationalize the numerator;

$$\Rightarrow \lim_{x \to 0} \frac{(\sqrt{x^2+1} - 1)}{x^2} \times \frac{(\sqrt{x^2+1} + 1)}{(\sqrt{x^2+1} + 1)} = \lim_{x \to 0} \frac{(x^2+1-1)}{x^2\sqrt{x^2+1} + x^2}$$

$$= \lim_{x \to 0} \frac{x^2}{x^2(\sqrt{x^2+1} + 1)} = \lim_{x \to 0} \frac{1}{\sqrt{x^2+1} + 1} = \frac{1}{2}$$

Note: The above example and that of ① and ③ satisfy the conditions for the use of the L'Hopital's rule, which we will discuss in a later section.

DIFFERENTIAL CALCULUS

(A) Differentiation By First Principle.

Defn:

The derivative of a given fn say $f(x)$ by first principle is given as:

$$f'(x) = \lim_{x \to h} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

* Some Notations of Differentiation

Let $y = f(x)$

$\Rightarrow$ the derivative of y is $f'(x) = \frac{dy}{dx} = y'$ etc.

Examples:

(1) Find the derivatives of the following:

(a) $f(x) = x^2 - 1$ (b) $f(x) = 10x + 3$.

Soln:

(a) using $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

where $f(x) = x^2 - 1$

$\Rightarrow f(x+h) = (x+h)^2 - 1 = x^2 + 2xh + h^2 - 1$

$$\Rightarrow f'(x) = \lim_{h \to 0} \frac{[x^2 + 2xh + h^2 - 1] - [x^2 - 1]}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - 1 - x^2 + 1}{h}$$

$$= \lim_{h \to 0} \frac{h(2x+h)}{h} = \lim_{h \to 0} 2x + h$$

$$= 2x$$

(b) Where $f(x) = 10x + 3 \Rightarrow f(x+h) = 10x + 10h + 3$

Hence, $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \frac{10x + 10h + 3 - (10x+3)}{h}$

$= \lim_{h \to 0} \frac{10h}{h} = \lim_{h \to 0} 10 = 10$

(B) TECHNIQUES OF DIFFERENTIATION (Standard Derivatives)

FUNCTION (y)	DERIVATIVE (dy/dx).
(i) ax^n	anx^{n-1}
(ii) $y = c$	0
(iii) $y = \ln x$	$1/x$
(iv) $y = \sin x$	$\cos x$
(v) $y = \cos x$	$-\sin x$
(vi) $y = \tan x$	$\sec^2 x$
(vii) $y = e^x$	e^x
(viii) $y = \operatorname{cosec} x$	
(ix) $y = \sec x$	
(x) $y = \cot x$	

RULES OF DIFFERENTIATION.

(a) The PRODUCT RULE: Let $y = uv$, where $u = f(x)$ & $v = f(x)$

then,

$$\frac{dy}{dx} = \frac{d(uv)}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Eg①: Differentiate $y = x^2 \sin x$ wrt x.

Soln:

given that $y = x^2 \sin x$, let $u = x^2$ and $v = \sin x$

$\Rightarrow \frac{du}{dx} = 2x$ and $\frac{dv}{dx} = \cos x$

$\rightarrow \frac{dy}{dx} = x^2(\cos x) + \sin x(2x)$

$= x^2 \cos x + 2x \sin x$

(b) The QUOTIENT RULE: Let $y = u/v$ where $u = f(x)$ & $v = f(x)$

$\Rightarrow \frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Eg(2) Differentiate $y = \frac{\ln x}{x+1}$

Soln:

$u = \ln x$ and $v = x+1$

$\Rightarrow \frac{du}{dx} = 1/x$ and $\frac{dv}{dx} = 1$

Hence; $\frac{dy}{dx} = \frac{(x+1)\frac{1}{x} - \ln x(1)}{}$

$= 1 + 1/x - \ln x$

(c) $\frac{d[u \pm v]}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Eg(3): Differentiate $y = 4x^3 - x^2 + 11x + 7$ wrt x.

Soln:

$\frac{dy}{dx} = 12x^{3-1} - 2x^{2-1} + 11x^{1-1} + 0$

$= 12x^2 - 2x + 11$

(d) The CHAIN RULE:

Given that (i) $y = f(g(x))$ or (ii) $y = f[g(h(x))]$

then if we set $g(x) = u$ and $h(x) = v$

$\Rightarrow$ (i) $y = f(u)$ and (ii) $y = f[u(v(x))]$.

Hence; For (i) $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

and (iii) $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$.

Eg (3): Differentiate the following fns:

(a) $y = \sin(3x+1)$ (b) $y = \sqrt{x^3 - 2x}$ (c) $y = \sqrt{\cos(5x-2)}$

Soln:

(a) $y = \sin(3x+1)$ Let $u = 3x+1$

$y = \sin u$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot 3$

$= 3\cos u = 3\cos(3x+1)$

(b) $y = \sqrt{x^3 - 2x}$. Let $u = x^3 - 2x \Rightarrow y = \sqrt{u} = u^{1/2}$

Then;

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= \frac{1}{2}u^{-1/2} \cdot 3x^2 - 2 = \frac{1}{2u^{1/2}} \cdot 3x^2 - 2$

$= \frac{3x^2 - 2}{2\sqrt{u}}$

(c) For $y = \sqrt{\cos(5x-2)}$. Let $u = \cos(5x-2)$ and $v = 5x-2$

$\Rightarrow u = \cos v$ and $y = \sqrt{u} = u^{1/2}$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$

$= \frac{1}{2}u^{-1/2} \cdot (-\sin v) \cdot 5$

$= -\frac{5}{2}u^{-1/2}\sin v = \frac{-5}{2} \cdot \frac{1}{u^{1/2}} \sin(5x-2)$

$= \frac{-5\sin(5x-2)}{2\sqrt{u}} = \frac{-5\sin(5x-2)}{2\sqrt{\cos(5x-2)}}$

~~IMPLICIT~~

DERIVATIVES OF LOGARITHMIC & EXPONENTIAL FUNCTIONS

Given that (a) $y = \ln x$, then $y' = 1/x$

(b) $y = a^x$, then $y' = a^x \ln a$

(c) $y = e^x$, then $y' = e^x$.

Examples

Differentiate the following

(1) $y = e^{5x-2}$ (2) $\ln(\sin x)$ (3) $e^{\ln x}$

Soln:

(1) Let $u = 5x-2$; $y = e^u$

using $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot (5)$

$= 5e^u = 5e^{5x-2}$

(2) Let $u = \sin x \Rightarrow y = \ln u$

using $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{u} \cdot \cos x$

$= \frac{\cos x}{\sin x} = \cot x$

(3) Let $u = \ln x \Rightarrow y = e^u$

using $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot \left(\frac{1}{x}\right)$

$= \frac{e^u}{x} = \frac{e^{\ln x}}{x}$

IMPLICIT DIFFERENTIATION

This kind of differentiation is applied when the fn does not clearly show which of the variables is dependent and the independent one.

Examples of implicit fns: $x^2 + y^2 = 3xy$, $x - y^3 = 2$

Note that;

Implicit fns are differentiated implicitly.

Examples

① If $x^2 + y^2 = y - 3x$, Find dy/dx.

Soln:

$x^2 + y^2 = y - 3x$

differentiate every term with respect to x;

$$2x + 2y\frac{dy}{dx} = \frac{dy}{dx} - 3$$

collect like terms;

$$2y\frac{dy}{dx} - \frac{dy}{dx} = -2x - 3$$

Factorize dy/dx

$$\frac{dy}{dx}(2y - 1) = -2x - 3$$

$$\therefore \frac{dy}{dx} = \frac{-2x - 3}{2y - 1}$$

② $x^2 + y^2 = 4xy$, find dy/dx.

Soln

$$2x^2 + 2y\frac{dy}{dx} = 4x\left(\frac{dy}{dx}\right) + y(4)$$ (applying product rule where $u = 4x$, $v = y$.)

$$2x^2 + 2y\frac{dy}{dx} = 4x\frac{dy}{dx} + 4y$$

$$2y\frac{dy}{dx} - 4x\frac{dy}{dx} = 4y - 2x^2$$

$$\frac{dy}{dx}(2y - 4x) = 4y - 2x^2$$

$$\therefore \frac{dy}{dx} = \frac{2(2y - x^2)}{2(y - 2x)} = \frac{2y - x^2}{y - 2x}$$

③ If $y = x^{x-1}$, find dy/dx

Soln:

take the log of both sides

$\ln y = \ln x^{x-1} \rightarrow \ln y = (x-1)\ln x$

differentiate; $\frac{1}{y}\frac{dy}{dx} = (x-1)\frac{1}{x} + \ln x(1)$

$\Rightarrow \frac{1}{y}\frac{dy}{dx} = 1 + \ln x - \frac{1}{x}$

$\therefore \frac{dy}{dx} = y\left(1 + \ln x - \frac{1}{x}\right)$

$= x^{x-1}\left(1 + \ln x - 1/x\right)$

④ If $e^{x+y} = xy$, find dy/dx

Soln:

$e^{x+y} = xy$

Let $u = x+y \Rightarrow e^u = xy$

using product rule on the RHS and chain rule on the LHS

For LHS; $\frac{dy}{du}\cdot\frac{du}{dx} = e^u\cdot\left(1 + \frac{dy}{dx}\right) = e^{x+y}\left(1 + \frac{dy}{dx}\right)$

RHS; $x\frac{dy}{dx} + y(1) = x\frac{dy}{dx} + y$

$\Rightarrow e^{x+y}\left(1 + \frac{dy}{dx}\right) = x\frac{dy}{dx} + y$

$\Rightarrow e^{x+y} + e^{x+y}\frac{dy}{dx} = x\frac{dy}{dx} + y$

$\Rightarrow e^{x+y}\frac{dy}{dx} - x\frac{dy}{dx} = y - e^{x+y}$

$\Rightarrow \frac{dy}{dx}\left(e^{x+y} - x\right) = y - e^{x+y}$

$\therefore \frac{dy}{dx} = \frac{y - e^{x+y}}{e^{x+y} - x}$

⑤ If $x^2+y^2-6x-8y+16=0$, find dy/dx.

Soln:

Given $x^2+y^2-6x-8y+16=0$

$$2x+2y\frac{dy}{dx}-6-8\frac{dy}{dx}=0$$

$$2y\frac{dy}{dx}-8\frac{dy}{dx}=6-2x$$

$$\frac{dy}{dx}(2y-8)=6-2x$$

$$\therefore \frac{dy}{dx}=\frac{6-2x}{2y-8}=\frac{3-x}{y-4}$$

HIGHER DERIVATIVES

$y'=\frac{dy}{dx}$ = first derivative i.e differentiating y once.

we have $y''=\frac{d^2y}{dx^2}$ → differentiating twice

$y'''=\frac{d^3y}{dx^3}$ → thrice etc.

Examples

① If $y=\sin(3x+1)$, $\frac{d^2y}{dx^2}=?$

Soln:

$y=\sin(3x+1)$

$$\frac{dy}{dx}=3\cos(3x+1)$$

$$\Rightarrow \frac{d^2y}{dx^2}=9(-\sin(3x+1))$$

$$=-9\sin(3x+1)$$

② If $y = 5x^3 - 2x^2 + 3$, find $\frac{d^3y}{dx^3}$.

Soln:

$\frac{dy}{dx} = 15x^2 - 4x$, $\frac{d^2y}{dx^2} = 30x - 4$

$\therefore \frac{d^3y}{dx^3} = 30$

APPLICATIONS OF DIFFERENTIATION

④ The L'Hopital's Rule: This is used to find the limits of rational fns that satisfy some conditions.

Statement: If $f(x)$ and $g(x)$ are differentiable fns and

$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$ (by direct substitution)

then, $\boxed{\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}}$ (which means, we shud different up and down, then substitute.)

N/B:

If $\lim_{x \to a} \frac{f'(x)}{g''(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$ then $\boxed{\lim_{x \to a} \frac{f'(x)}{g'(x)} = \lim_{x \to a} \frac{f''(x)}{g''(x)}}$

Examples:

Evaluate the following limits

① $\lim_{x \to 0} \frac{\sin x}{x}$ ② $\lim_{x \to 0} \frac{e^x - x - 1}{x^2}$ ③ $\lim_{x \to -2} \frac{x+2}{x^2+3x+2}$

④ $\lim_{x \to \infty} \frac{x}{x+1}$

Soln:

① $\lim_{x \to 0} \frac{\sin x}{x} = \frac{0}{0}$ (by direct substitution).

$\Rightarrow \lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \frac{\cos x}{1}$. (differentiating up and down).

$= \frac{\cos 0}{1} = \frac{1}{1} = 1$

② $\lim_{x\to 0} \frac{e^x - x - 1}{x^2} = \frac{e^0 - 0 - 1}{0^2} = \frac{0}{0}$ (direct substitution)

$\Rightarrow \lim_{x\to 0} \frac{e^x - x - 1}{x^2} = \lim_{x\to 0} \frac{e^x - 1}{2x} = \frac{0}{0}$ (still 0/0 after 1st differentiation)

$= \lim_{x\to 0} \frac{e^x}{2} = \frac{e^0}{2} = \frac{1}{2}$

③ $\lim_{x\to -2} \frac{x+2}{x^2+3x+2} = \frac{0}{0}$ (by direct sub.)

$= \lim_{x\to -2} \frac{1}{2x+3} = \frac{1}{-4+3} = \frac{-1}{1}$

④ $\lim_{x\to \infty} \frac{x}{x+1} = \frac{\infty}{\infty}$ (by direct sub.)

$= \lim_{x\to \infty} \frac{1}{1} = \frac{1}{1}$

LIMITS TO INFINITY.

To evaluate an algebraic fn in a limit to infinity, we divide each term present by the variable with the highest degree ~~pre~~ in the fn. It is usually applicable with rational algebraic fn.

Examples:

Evaluate the following limits

① $\lim_{x\to\infty} \frac{x+3}{2x-5}$ ② $\lim_{x\to\infty} \frac{3x^2-4x}{5x^2}$ ③ $\lim_{x\to\infty} \sqrt{\frac{x^2-3}{x^3+6x-1}}$

Sols:

① $\lim_{x\to\infty} \frac{x+3}{2x-5}$

divide every term by x

$$\Rightarrow \lim_{x\to\infty} \frac{1+3/x}{2-5/x}$$

{N/B: $x+\infty=\infty$; $\infty-x=\infty$; $\infty\times x=\infty$; $\infty\div x=\infty$; $x\div\infty=0$.}

$$= \frac{1+3/\infty}{2-5/\infty} = \frac{1+0}{2-0}$$

$$= \underline{\underline{1/2}}$$

(2) $$\lim_{x\to\infty} \frac{3x^2-4x}{5x^2} = \lim_{x\to\infty} \frac{3-4/x}{5}$$

$$= \underline{\underline{3/5}}$$

(3) $$\lim_{x\to\infty} \sqrt{\frac{x^2-3}{x^3+6x+1}} = \sqrt{\lim_{x\to\infty} \frac{1/x-3/x^3}{1+6/x^2+1/x^3}}$$

$$= \sqrt{\frac{1/\infty - 3/\infty}{1+6/\infty+1/\infty}} = \sqrt{\frac{0-0}{1}} = \sqrt{0} = \underline{\underline{0}}$$

Alternatively;

If given $\lim_{x\to\infty} \frac{ax^n}{bx^n} = \frac{a}{b}$,

⇒ Get the coefficients of the variable with highest degree in the numerator and denominator; then divide them.

Example

(a) From $\lim_{x\to\infty} \frac{x+3}{2x-5}$, the ~~variable~~ term with the highest degree is x

$\therefore \lim_{x\to\infty} \frac{x+3}{2x-5} = \underline{\underline{\frac{1}{2}}}$ {since the coefficient of x in the numerator is 1 and that of the denominator is 2.}

(b) $\lim_{x \to \infty} \frac{7+2x^2-5x^3}{10x^2+x^3} = \frac{+5}{1} = -5$

(c) $\lim_{x \to \infty} \frac{7+2x^2}{10x^2+x^3} = \frac{0}{1}$ { Since the coefficient of x^3 in the numerator is 0.

$= 0$

(B) TANGENT & NORMAL TO A CURVE.

Assuming that $Q(x_0, y_0)$ is a point on a curve, $y = f(x)$, then the tangent of the curve at the point, Q is given as;

$$\text{tangent}[\text{at } (x_0, y_0)] = \left.\frac{dy}{dx}\right|_{x=x_0}$$

It is also called the SLOPE of the curve. Let us represent that as m_t. Meanwhile, the NORMAL to a curve, $y = f(x)$ is given as!

$$m_n = -\frac{1}{m_t}$$

, this is because, the tangent to a curve is always perpendicular to the normal of the curve.

Examples

Find the tangent and normal of each of the following fns!

(1) $y = 7+2x^2-5x^3$ at $(1, -2)$

(2) $2x^3 + y^2 \neq 3xy$ at the point $(+3, 0)$

Soln:

(1)(a) $y = 7+2x^2-5x^3$

$\rightarrow \frac{dy}{dx} = 4x - 15x^2$

$\Rightarrow \left.\frac{dy}{dx}\right|_{(1,-2)} = 4(1) - 15(1)^2 = -11$ (Substitute $x=1$).

$\therefore$ tangent of the fn at $(1,-2) = -11$

(b) For the Normal at the pt $(1,-2)$

$$m_n = -\frac{1}{m_t} = \frac{1}{-11}$$

$$= \frac{1}{11}$$

(2) $2x^3 + y^2 = 3xy$ at $(-3,0)$.

Soln:

(a) Differentiating Implicitly;

$$6x^2 + 2y\frac{dy}{dx} = 3x\frac{dy}{dx} + y(3)$$

$$2y\frac{dy}{dx} - 3x\frac{dy}{dx} = 3y - 6x^2$$

$$\frac{dy}{dx} = \frac{3y - 6x^2}{2y - 3x}$$

$\therefore \left.\frac{dy}{dx}\right|_{(-3,0)} = \frac{3(-3) - 6(-3)^2}{2(-3) - 3(-3)} = \frac{-9 - 54}{-6 + 9}$

$$= \frac{-63}{3} = -21$$

(b) Normal, $m_n = -\frac{1}{m_t} = -\frac{1}{-21}$

$$= \frac{1}{21}$$

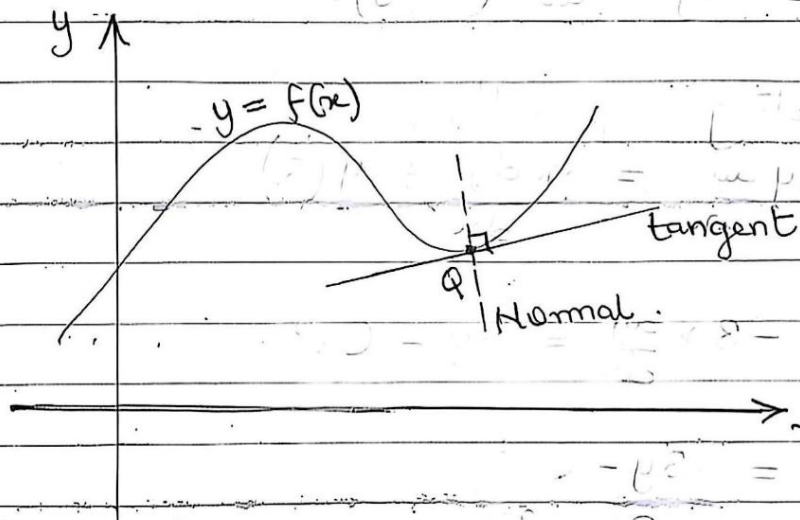
EQUATION OF TANGENT & NORMAL TO A CURVE.

(A) Given a point $Q(x_0, y_0)$ on a curve $y = f(x)$ with tangent as $\left.\frac{dy}{dx}\right|_{x=x_0}$, then the eqn of the tangent to the curve at the point $Q(x_0, y_0)$ is given by:

$$\boxed{y - y_0 = m_t(x - x_0)}$$ where - m_t = tangent of curve.

(B) The eqn of the normal to a curve, $y = f(x)$ is given by:

$$\boxed{y - y_0 = m_n(x - x_0)}$$ where $m_n = \frac{-1}{m_t}$.



Examples: ① Find the eqn of the tangent to the curve $2x^3 + y^2 = 3xy$ at the point $(1, -2)$.

Soln:

Using implicit differentiation;

$$6x^2 + 2y\frac{dy}{dx} = 3x\frac{dy}{dx} + 3y$$

$$\Rightarrow (2y - 3x)\frac{dy}{dx} = 3y - 6x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{3y - 6x^2}{2y - 3x} \quad \text{and} \quad \left.\frac{dy}{dx}\right|_{(1,-2)} = \frac{3(-2) - 6(1)}{2(-2) - 3(1)}$$

$$\Rightarrow m_t = \frac{12}{7}$$

∴ Eqn of tangent line becomes:

$$y+2 = \frac{12}{7}(x-1)$$

$$\Rightarrow 7y+14 = 12(x-1)$$

$$\Rightarrow 7y+14 = 12x-12$$

$$\Rightarrow 7y-12x+26 = 0$$

② Find the slope of the normal line to the curve $y=2x^3-5x+1$ at the pt whose x-coordinate is 1.

Soln:

$$\frac{dy}{dx} = 6x^2-5$$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{x=1} = m_t = 6(1)-5 = 1$$

Hence, $m_n = \frac{-1}{m_t} = \underline{\underline{-1}}$

③ Find the eqn of the normal line in example ② above.

Soln:

using: $y-y_1 = m_n(x-x_1)$

where $x_1=1$, $y_1 = 2(1)^3-5(1)+1$

$= -2$

$\Rightarrow y-(-2) = -1(x-1)$

$\Rightarrow y+2 = -x+1$

∴ the eqn is: $\underline{\underline{y+x+1=0}}$

STATIONARY POINTS.

Also called turning points, there are 3 of them:

(a) Maximum pt
(b) Minimum pt
(c) Inflexion pt

To get the stationary points of a fn, we differentiate, and equate to zero and solve for x.

Examples

① Find the stationary points of the fn, $f(x) = x^3-3x+1$.

Soln:

$f'(x) = 3x^2-3$

$\Rightarrow 3x^2-3 = 0$

$\Rightarrow x^2-1 = 0$

$\Rightarrow (x-1)(x+1) = 0$

$\therefore x = 1$ or -1

⇒ The turning pts are $\underline{\underline{1 \text{ & } -1}}$

N/B: To determine the type of turning pt, we differentiate $f(x)$ twice and substitute the x-values. If

(i) $f''(x) > 0 \Rightarrow x$ is minimum
(ii) $f''(x) < 0 \Rightarrow x$ is maximum
(iii) $f''(x) = 0 \Rightarrow x$ is inflexion.

Eg ②: Determine the nature of the stationary pts. for the fn $y = x^3-x^2-5x+1$

Soln:

$$\frac{dy}{dx} = 3x^2-2x-5$$

$\Rightarrow 3x(x+1) - 5(x+1) = 0$

$\Rightarrow 3x - 5 = 0$ or $x+1 = 0$

$\Rightarrow x = 5/3$ or -1

To test the Nature;

$f''(x) = 6x - 2$

(i) $f''(1) = 6(1) - 2 = 4 > 0$

$\therefore x = -1$ is maximum pt

(ii) $f''(5/3) = 6(5/3) - 2 = 8 > 0$

$\Rightarrow x = 5/3$ is a minimum pt

(3) Determine the minimum pt and minimum value for the fn $f(x) = x^3 - 3x + 1$.

Sol$\bar{n}$:

(a) $f'(x) = 3x^2 - 3 = 0$

$\Rightarrow x^2 - 1 = 0$

$\therefore x = 1$ or -1

and

$f''(x) = 6x$

$f''(1) = 6 > 0$

$\therefore x = 1$ is minimum

(b) For minimum value, substitute the minimum pt into $f(x)$.

→ Minimum Value $= f(1)$

$= 1^3 - 3(1) + 1$

$= -1$

Exercise:

Find the stationary pts of the fn $y = 4x^3 + 15x^2 - 18x + 7$ and also determine their nature.

⑥ INTEGRATION.

* Integration as anti-derivative
* Integral Calculus Theorems
* Integrals Leading to logarithms
* Integration by ~~fra~~ Substitution
* Integration by parts
* ✓ of trig functions with even and odd powers
* Integration of exponential fns
* Applications of Integration.

(I) INTEGRATION AS ANTI-DERIVATIVE

We denote integration of a fn $f(x)$ by

$$\int f(x)\,dx$$

⇒ Integral of $f(x)$ with respect to x.. Where $\int$ is called Integral and $f(x)$ is called the Integrand.

Note: The process of evaluating $\int f(x)\,dx$ is called INTEGRATION, and it would usually give another fn of x and a CONSTANT of integration c.

i.e. $$\int f(x)\,dx = F(x) + c$$

Examples:

① $\int 2x\,dx = x^2 + c$

② $\int (6x+1)\,dx = 3x^2 + x + c$

③ $\int (3x^2 - x + 5)\,dx = x^3 - \frac{x^2}{2} + 5x + c$

④ $\int \cos x\,dx = \sin x + c$

⑤ $\int e^x\,dx = e^x + c$ etc.

Formulas:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$$

Basic Theorems

Let $f(x)$ and $g(x)$ be fns that are integrable, then:

(a) $\int k f(x)\,dx = k\int f(x)\,dx$

e.g: $\int 2x\,dx = 2\int x^1 dx$

$= x\left(\frac{x^{1+1}}{1+1}\right) + c$

$= x^2 + c$

(b) $\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$

⇒ If there are more one terms in a fn, you integrate them one after another.

Examples

Evaluate the following integrals.

① $\int (3x^2 + 5x + 2)\,dx$

$= \frac{3x^3}{3} + \frac{5x^2}{2} + \frac{2x^1}{1} + c$

$= x^3 + \frac{5x^2}{2} + 2x + c$

② $\int (4x^3 + 6x^2 + 4x - 3)\,dx$

$= \frac{4x^4}{4} + \frac{6x^3}{3} + \frac{4x^2}{2} - 3x + c$

$= x^4 + 2x^3 + 2x^2 - 3x + c$

③ $\int \frac{4}{x^2}\,dx = \int 4x^{-2}\,dx$

$= \frac{4x^{-1}}{-1} + c.$

$= -\frac{4}{x} + c$

④ $\int \frac{x^3 - 2x^2 + 4x}{x}\,dx = \int (x^2 - 2x + 4)\,dx$

$= \frac{x^3}{3} - x^2 + 4x + c$

(II) STANDARD INTEGRALS

$f(x)$	$\int f(x)\,dx$
① x^n	$x^{n+1}/n+1 + c$
② k	kx
③ $\sin x$	$-\cos x + c$
④ $\cos x$	$\sin x + c$
⑤ $\sec^2 x$	$\tan x + c$
⑥ e^x	$e^x + c$
⑦ $1/x$	$\ln x + c$
⑧ $a^x \ln a$	$a^x + c$
⑨ $\sin(ax)$	$-\frac{\cos(ax)}{a} + c$
⑩ $\cos(ax)$	$\frac{\sin(ax)}{a} + c$
⑪ $\sec^2(ax)$	$\frac{\tan(ax)}{a} + c$
⑫ e^{ax}	$\frac{e^{ax}}{a} + c$
⑬ $\frac{f'(x)}{f(x)}$	$\ln[f(x)] + c$
⑭ a^x	$\frac{a^x}{\ln a} + c$

(III) Integration Leading To Log.

Any integral that has its numerator as a derivative of the denominator gives the ln of the denominator i.e.

$$\int \frac{f'(x)}{f(x)}\,dx = \ln(f(x)) + c$$

Examples

① $\int \frac{1}{x}\,dx = \ln x + c$

(since the derivative of x is 1.)

② $\int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx$

$= \ln(\sin x) + c$

③ $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

$= -\int \frac{-\sin x}{\cos x}\,dx$

$= -\ln(\cos x) + c$

$= \ln(\sec x) + c$

④ $\int \frac{2x}{x^2+1}\,dx = \ln(x^2+1) + c$

⑤ $\int \frac{1}{3x-5}\,dx = \frac{1}{3}\int \frac{1}{x - 5/3}\,dx$

$= \frac{1}{3}\ln(x - 5/3) + c$

OR

$\int \frac{1}{3x-5}\,dx = \frac{1}{3}\int \frac{3}{3x-5}\,dx$

$= \frac{1}{3}\ln(3x-5) + c$

N/B: The second approach is preferable

⑥ $\int \frac{3}{2x+3}\,dx = \frac{3}{2}\int \frac{2}{2x+3}\,dx$

$= \frac{3}{2}\ln(2x+3) + c$

⑦ $\int \frac{4x-8}{x^2-4x+5}\,dx = 2\int \frac{2x-4}{x^2-4x+5}\,dx$

$= 2\ln(x^2-4x+5) + c$

(IV) Integration By Partial Fraction

This involves fns that will be simplified via partial fractions and then apply the integration approach that leads to logarithms.

Examples

(i) $\int \frac{4}{x^2-9}\,dx = \int \frac{4}{(x+3)(x-3)}\,dx$

$= \int \left[\frac{2/3}{x+3} - \frac{2/3}{x+3}\right] dx$

$= \frac{2}{3}\int \left(\frac{1}{x-3} - \frac{1}{x+3}\right) dx.$

$= \frac{2}{3}\left[\ln(x-3) - \ln(x+3)\right] + c$

$= \frac{2}{3}\ln\left(\frac{x-3}{x+3}\right) + c$

(2) $\int \frac{5x+2}{x^2+3x+2}\,dx$

By partial fractions;

$\frac{5x+2}{x^2+3x+2} = \frac{8}{x+2} - \frac{3}{x+1}$

$\Rightarrow \int \frac{5x+2}{x^2+3x+2}\,dx = \int \left(\frac{8}{x+2} - \frac{3}{x+1}\right) dx$

$= \int \frac{8}{x+2}\,dx - \int \frac{3}{x+1}\,dx$

$= 8\int \frac{1}{x+2}\,dx - 3\int \frac{1}{x+1}\,dx$

$= 8\ln(x+2) - 3\ln(x+1) + c$

OR

$= \ln(x+2)^8 - \ln(x+1)^3 + c$

$= \ln\left[\frac{(x+2)^8}{(x+1)^3}\right] + c$

(3) $\int \frac{1}{x^2-1}\,dx$

By partial fractions;

$\frac{1}{x^2-1} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$

$\Rightarrow \int \frac{1}{x^2-1}\,dx = \frac{1}{2}\int \left(\frac{1}{x-1} - \frac{1}{x+1}\right) dx$

$= \frac{1}{2}\left[\ln(x-1) - \ln(x+1)\right] + c$

$= \frac{1}{2}\ln\left(\frac{x-1}{x+1}\right) + c$

(4) $\int \frac{3x+5}{(x+2)(x-1)}\,dx$

By Partial Fractions;

$\frac{3x+5}{(x+2)(x-1)} = \frac{1}{3(x+2)} + \frac{8}{3(x-1)}$

$\Rightarrow \int \frac{3x+5}{(x+2)(x-1)}\,dx = \int \left[\frac{1}{3(x+2)} + \frac{8}{3(x-1)}\right] dx$

$= \frac{1}{3}\int \frac{1}{x+2}\,dx + \frac{8}{3}\int \frac{1}{x-1}\,dx$

$= \frac{1}{3}\ln(x+2) + \frac{8}{3}\ln(x-1) + c$

(5) $\int \frac{2x+1}{x^2+x-6}\,dx$

Factor the denominator;

$x^2+x-6 = (x-2)(x+3)$

By partial fractions;

$\frac{2x+1}{x^2+x-6} = \frac{2x+1}{(x-2)(x+3)}$

$= \frac{1}{x-2} + \frac{1}{x+3}$

Exercises

Evaluate the following.

(1) $\int \frac{5}{x^2-4}\,dx$ (2) $\int \frac{1}{x^2+2x}\,dx$

(3) $\int \frac{4x+1}{2x^2+x+3}\,dx$ (4) $\int \frac{3-x}{x^2-9}\,dx$

(5) $\int \frac{1}{5x+2}\,dx$ (6) $\int \frac{x+1}{x^2+x}\,dx$

(V) INTEGRATION BY SUBSTITUTION.

This is the analogue of Chain rule method in differentiation

Examples

(1) $\int (2x+1)^5\,dx$

Solⁿ: Let $u = 2x+1 \Rightarrow du = 2dx$

Hence, $\Rightarrow dx = \frac{du}{2}$

$$\int (2x+1)^5\,dx = \int u^5 \cdot \frac{du}{2}$$

$$= \frac{1}{2}\int u^5\,du = \frac{1}{2}\frac{u^6}{6} + c$$

$$= \frac{(2x+10)^6}{12} + c$$

(2) $\int x(x^2+1)^3\,dx$ Let $u = x^2+1$

$\Rightarrow du = 2x\,dx \Rightarrow x\,dx = \frac{du}{2}$

Hence;

$$\int x(x^2+1)^3\,dx = \int u^3 \frac{du}{2}.$$

$$= \frac{1}{2}\int u^3\,du = \frac{1}{2}\frac{u^4}{4} + c$$

$$= \frac{(x^2+1)^4}{8} + c$$

(3) $\int \frac{dx}{(3x+4)^2} = \int \frac{1}{(3x+4)^2}\,dx$

Let $u = 3x+4 \Rightarrow du = 3\,dx$

$\Rightarrow dx = du/3$

Hence; $\int \frac{dx}{(3x+4)^2} = \int \frac{1}{u^2} \cdot \frac{du}{3}$

$$= \frac{1}{3}\int u^{-2}\,du = \frac{1}{3}\frac{u^{-1}}{-1} + c$$

$$= \frac{-1}{3(3x+4)} + c$$

(4) $\int \cos(2x)\,dx \Rightarrow u = 2x \Rightarrow dx = \frac{du}{2}$

$$\Rightarrow \int \cos(2x)\,dx = \int \cos u \cdot \frac{du}{2}$$

$$= \frac{1}{2}\int \cos u\,du = \frac{1}{2}\sin u + c$$

$$= \frac{1}{2}\sin(2x) + c$$

(5) $\int \frac{x\,dx}{\sqrt{x^2+4}}$

Let $u = x^2+4 \Rightarrow du = 2x\,dx \Rightarrow x\,dx = \frac{du}{2}$

$$\Rightarrow \int \frac{x\,dx}{\sqrt{x^2+4}} = \int \frac{1}{u^{1/2}}\frac{du}{2} = \frac{1}{2}\int u^{-1/2}\,du.$$

$$= \frac{1}{2} \cdot \frac{u^{1/2}}{1/2} + c$$

$$= \sqrt{u} + c = \sqrt{x^2+4} + c$$

(6) $\int \frac{e^{2x}}{1+e^{2x}}\,dx$; Let $u = 1+e^{2x}$

$\Rightarrow du = 2e^{2x}\,dx$

$\Rightarrow dx = du/2e^{2x}$.

$$\Rightarrow \int \frac{e^{2x}}{1+e^{2x}}\,dx = \int \frac{e^{2x}}{u} \cdot \frac{du}{2e^{2x}}$$

$$= \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln u + c$$

$$= \frac{1}{2}\ln(1+e^{2x}) + c$$

EXERCISES

① $\int \frac{3x^2}{(x^3+1)^2}\,dx$ ② $\int \frac{dx}{(2x+3)^4}$

③ $\int \frac{x\,dx}{(x^2+1)^2}$ ④ $\int \frac{1}{x\ln x}\,dx$

⑤ $\int \frac{\sin(\ln x)}{x}\,dx$ ⑥ $\int \frac{x}{x^2+9}\,dx$

Solution

⑥ INTEGRATION OF TRIG FUNCTIONS

* $\int \sin x\,dx = -\cos x + c$
* $\int \cos x\,dx = \sin x + c$
* $\int \sec^2 x\,dx = \tan x + c$
* $\int \operatorname{cosec}^2 x\,dx = -\cot x + c$
* $\int \sec x \tan x\,dx = \sec x$

① $\int \sin^2(x)\,dx$

where $\sin^2 x = \frac{1-\cos 2x}{2} = \frac{1}{2} - \frac{1}{2}\cos 2x$

$\Rightarrow \int \sin^2 x\,dx = \int \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) dx$

$= \frac{x}{2} - \frac{\sin 2x}{4} + c$

② $\int \cos^2 x\,dx$

where $\cos^2 x = \frac{1+\cos(2x)}{2}$

$\Rightarrow \int \cos^2 x\,dx = \int \left(\frac{1}{2} + \frac{1}{2}\cos(2x)\right) dx$

$= \frac{x}{2} + \frac{\sin(2x)}{4} + c$

③ $\int \sin x \cos x\,dx$ (from compound angles)

where $\sin x \cos x = \frac{1}{2}\sin(2x)$

$\Rightarrow \int \sin x \cos x\,dx = \int \frac{1}{2}\sin(2x)\,dx$

$= \frac{1}{2}\left(-\frac{\cos(2x)}{2}\right) + c$

$= -\frac{\cos(2x)}{4} + c$

④ $\int \sec x\,dx$

multiply by $\frac{\sec x + \tan x}{\sec x + \tan x}$

$= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x}\right) dx$

$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\, dx$

Let

$u = \sec x + \tan x$

$\Rightarrow \frac{du}{dx} = \sec x \tan x + \sec^2 x$

$\Rightarrow$ the derivative of the denom. gives the numerator.

$\therefore \int \sec x\,dx = \ln(\sec x + \tan x) + c$

(VII) INTEGRATION OF EXP. FUNCTION

$\int e^x dx = e^x + c$	$\int a^x \ln a\, dx = a^x + c$
$\int e^{ax} dx = \frac{e^{ax}}{a} + c$	$\int a^x dx = \frac{a^x}{\ln a} + c$

* In general, let $f(x) = e^{g(x)}$

$$\int e^{g(x)} dx = \frac{e^{g(x)}}{g'(x)} + c$$

Examples

① $\int e^{3x} dx = \frac{e^{3x}}{3} + c$

$= \frac{1}{3} e^{3x} + c$

② $\int e^{-2x} dx$ — Let $u = -2x$

$\frac{du}{dx} = -2$

$\Rightarrow \int e^{-2x} dx = \frac{e^{-2x}}{-2} + c$

$= -\frac{1}{2} e^{-2x} + c$

③ $\int (2e^{4x} - 3e^{2x}) dx$

$= \frac{2e^{4x}}{4} - \frac{3e^{2x}}{2} + c$

$= \frac{1}{2} e^{4x} - \frac{3}{2} e^{x} + c$

$= \frac{1}{2}(e^{4x} - 3e^{x}) + c$

④ $\int e^{(x+1)} dx = \frac{e^{x+1}}{1} + c$

$= e^{x+1} + c$

⑤ $\int e^{(2x+3)} dx = \frac{e^{(2x+3)}}{(2x+3)'} + c$

$= \frac{e^{(2x+3)}}{2} + c$

⑥ $\int e^{\left(\frac{1-5x}{2}\right)} dx$

$= \int e^{\left(\frac{1}{2} - \frac{5x}{2}\right)} dx$

$= \frac{e^{\left(\frac{1}{2} - \frac{5x}{2}\right)}}{\left(\frac{1}{2} - \frac{5x}{2}\right)'} + c$

$= \frac{e^{\left(\frac{1-5x}{2}\right)}}{-5/2} + c$

$= -\frac{2e^{\left(\frac{1-5x}{2}\right)}}{5} + c$

(VIII) INTEGRATION BY PARTS

This is analogous of the product rule in differentiation. It is a technique used to evaluate integrals involving the product of 2 fns.

The formula is given by:

$$\boxed{\int u\,dv = uv - \int v\,du}$$

CHOOSING u and dv

The LIATE rule will be helpful:

L - Logarithmic ($\ln x$)
I - Inverse trig ($\tan^{-1}x$, $\sin^{-1}x$)
A - Algebraic (x^2 etc)
T - Trigonometric ($\sin x$, $\cos x$)
E - Exponential (e^x, 2^x).

N/B: Pick u from the higher category in the list, and let the rest be dv.

Examples

① $\int xe^x dx$

Let $u = x$ and $dv = e^x dx$

$\Rightarrow du = dx$ & $v = e^x$.

Substitute into the formula:

$\int u\,dv = uv - \int v\,du$

$\Rightarrow \int xe^x dx = xe^x - \int e^x dx$

$= xe^x - e^x + c$

$= e^x(x-1) + c$

② $\int x \ln x\, dx$

Let $u = \ln x \Rightarrow du = \frac{1}{x}dx$

$dv = x\,dx \Rightarrow v = x^2/2$

$\Rightarrow \int x\ln x\,dx = \ln x\left(\frac{x^2}{2}\right) - \int \frac{x^2}{2}\cdot\frac{1}{x}dx$

$= \frac{x^2}{2}\ln x - \frac{1}{2}\int x\,dx$

$= \frac{x^2}{2}\ln x - \frac{1}{2}\frac{x^2}{2} + c$

$= \frac{x^2}{2}\ln x - \frac{1}{4}x^2 + c$

③ $\int x\cos x\,dx$

Let $u = x \Rightarrow du = dx$.

$dv = \cos x\,dx \Rightarrow v = \sin x$

$\Rightarrow \int x\cos x\,dx = x\sin x - \int \sin x\,dx$

$= x\sin x - (-\cos x) + c$

$= x\sin x + \cos x + c$

④ $\int x^2 e^x dx$

Let $u = x^2 \Rightarrow du = 2x\,dx$

$dv = e^x dx \Rightarrow v = e^x$.

$\Rightarrow \int x^2e^x dx = x^2e^x - \int e^x \cdot 2x\,dx$

$= x^2e^x - \int 2xe^x dx$.

We apply integration by parts again, on $\int 2xe^x dx$

Let $u = 2x \Rightarrow du = 2dx$

$dv = e^x dx \Rightarrow v = e^x$

$\Rightarrow \int 2xe^x dx = 2xe^x - \int e^x \cdot 2dx$

$= 2xe^x - 2\int e^x dx$

$= 2xe^x - 2e^x + c$

Hence,

$\int x^2e^x dx = x^2e^x - [2xe^x - 2e^x] + c$

$= x^2e^x - 2xe^x + 2e^x + c$

⑤ $\int x\tan^{-1}x\,dx$

Let $u=\tan^{-1}x \rightarrow du=\frac{1}{1+x^2}dx$

$dv=x\,dx \Rightarrow v=\frac{x^2}{2}$

Hence;

$$\int x\tan^{-1}x\,dx=\frac{x^2}{2}\tan^{-1}x-\int\frac{x^2}{2}\cdot\frac{1}{1+x^2}dx$$

$$=\frac{x^2}{2}\tan^{-1}x-\frac{1}{2}\int\frac{x^2}{1+x^2}dx$$

$$=\frac{x^2}{2}\tan^{-1}x-\frac{1}{2}\int\left(1-\frac{1}{1+x^2}\right)dx$$

$$=\frac{x^2}{2}\tan^{-1}x-\frac{1}{2}\left[x-\tan^{-1}x\right]+C$$

$$=\frac{x^2}{2}\tan^{-1}x-\frac{x}{2}+\frac{1}{2}\tan^{-1}x+C$$

$$=\left(\frac{x^2+1}{2}\right)\tan^{-1}x-\frac{x}{2}+C$$

⑥ $\int e^x\cos x\,dx$.

Let $u=e^x \Rightarrow du=e^x dx$

$dv=\cos x\,dx \Rightarrow v=\sin x$

Hence;

$$\int e^x\cos x\,dx=e^x\sin x-\int\sin x\cdot e^x dx$$

We apply the rule on $\int e^x\sin x\,dx$

Let $u=e^x \Rightarrow du=e^x dx$

$dv=\sin x\,dx \Rightarrow v=-\cos x$

$$\int e^x\sin x\,dx=-e^x\cos x+\int e^x\cos x\,dx$$

Then;

$$\int e^x\cos x\,dx=e^x\sin x-\left[-e^x\cos x+\int e^x\cos x\,dx\right]$$

$$\int e^x\cos x\,dx=e^x\sin x+e^x\cos x-\int e^x\cos x\,dx$$

collect like terms;

$$2\int e^x\cos x\,dx=e^x(\sin x+\cos x)$$

$$\therefore\int e^x\cos x\,dx=\frac{e^x(\sin x+\cos x)}{2}+C$$

EXERCISES

① $\int x\sin x\,dx$ ② $\int \ln x\,dx$.

③ $\int x^2\ln x\,dx$ ④ $\int x^2\cos x\,dx$

⑤ $\int xe^{2x}dx$. ⑥ $\int e^x\sin x\,dx$.

ANSWERS (CHOOSE THE CORRECT)

① (A) $-x\cos x+\sin x+c$

(B) $x\cos x+\sin x+c$

(C) $-x\cos x+\cos x+C$

(D) $-x\cos x-\sin x+C$

② (A) $x\ln x+x+C$ (B) $x\ln x-x+C$

(C) $\ln^2 x+C$ (D) $\frac{1}{x}+C$

④ (A) $x^2\sin x+2x\cos x-2\sin x+C$

(B) $x^2\sin x-2x\cos x+2\sin x+C$

(C) $x^2\cos x-2x\sin x+2\cos x+C$

(D) $x^2\sin x-2x\cos x-2\sin x+C$

⑤ (A) $\frac{x}{2}e^{2x}-\frac{1}{4}e^{2x}+C$

(B) $\frac{x}{2}e^{2x}+\frac{1}{4}e^{2x}+C$

(C) $\frac{1}{2}xe^{2x}-\frac{1}{2}e^{2x}+C$

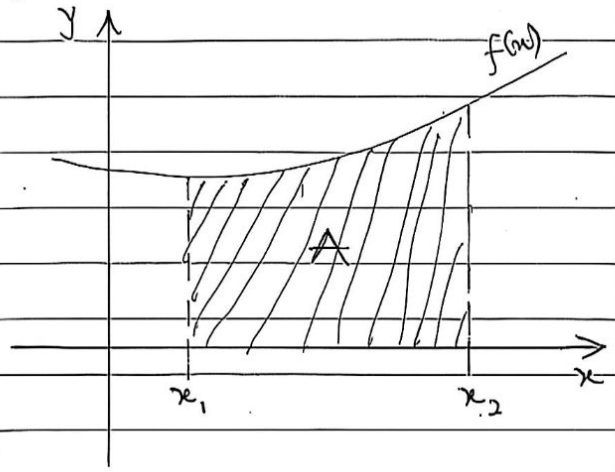
(D) $xe^{2x}+e^{2x}+C$

APPLICATIONS OF INTEGRATION

- Area Under a Curve
- Area Between Two Curves.

(A) AREA UNDER A CURVE

Let $y = f(x)$ be a curve, the area under y bounded by x, lines x_1 and x_2 shown in the diagram below;



is given by:

$$A = \int_{x_1}^{x_2} f(x)\,dx \text{ or } \int_{x_1}^{x_2} (f_1 - f_2)\,dx$$

(where f_1 – higher fn, f_2 – lower fn.)

Examples

① Find the area under the curve $y = x^2$ from $x = 0$ to $x = 2$.

Soln:

$$\int_{x_1}^{x_2} f(x)\,dx = \int_0^2 x^2\,dx = \frac{x^3}{3}\Big|_0^2$$

$$= \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}$$

② Find the area bounded by the curves $y = x^2$ and $y = 4$.

Soln:

$$A = \int_{x_1}^{x_2} (f_1(x) - f_2(x))\,dx$$

x_1 and x_2 will be the values of x where $y = 4$ and $y = x^2$ intersect.

i.e; $x^2 = 4 \Rightarrow x_1 = -2,\ x_2 = 2$

NB: Over the interval -2 to 2, $y = 4$ is above $y = x^2$,

Hence

$$A = \int_{-2}^{2} (4 - x^2)\,dx$$

$$= \left[4x - \frac{x^3}{3}\right]_{-2}^{2}$$

$$= \left(4(2) - \frac{2^3}{8}\right) - \left(4(-2) - \frac{(-2)^3}{8}\right)$$

$$= 8 - \frac{8}{3} + 8 - \frac{8}{3}$$

$$= 16 - \frac{16}{3}$$

$$\therefore \frac{48 - 16}{3} = \frac{32}{3}$$

③ Find the area between the curves $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$.

Soln: $x_1 = 0$ and $x_2 = \pi/2$.

Determine which fn is on top.

For $x \in [0, \pi/4)$: $\cos x > \sin x$

For $x \in (\pi/4, \pi/2]$: $\sin x > \cos x$

and at $x = \pi/4$; $\sin x = \cos x$.

So we split the integral:

$$A = \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx$$

$$= \left[\sin x + \cos x\right]_0^{\pi/4} + \left[-\cos x - \sin x\right]_{\pi/4}^{\pi/2}$$

$= (\sqrt{2}-1) + (\sqrt{2}-1)$

$= 2(\sqrt{2}-1)$

(4) Find the area of the region bounded below by $y = x^2 - 1$, above ~~below~~ by $y = -x$ and on the sides by $x = -1$ and $x = 1$.

Soln:

$$A = \int_{-1}^{1} [-x - (x^2 - 1)]\, dx$$

$$= \int_{-1}^{1} (-x - x^2 + 1)\, dx$$

$$= \left[-\frac{x^2}{2} - \frac{x^3}{3} + x \right]_{-1}^{1}$$

$$= \left(-\frac{1}{2} - \frac{1}{3} + \frac{1}{1}\right) - \left(-\frac{1}{2} + \frac{1}{3} - \frac{1}{1}\right)$$

$$= -\frac{1}{2} - \frac{1}{3} + 1 + \frac{1}{2} - \frac{1}{3} + 1$$

$$= 2 - 2\left(\frac{1}{3}\right) = 2 - \frac{2}{3}$$

$$= 1\frac{1}{3}$$

(5) Find the area bdd by the curve $y = x^2 + 3$, the x-axis and the ordinates $x = 1$ & $x = 3$.

Soln:

$$A = \int_{1}^{3} (x^2 + 3)\, dx \quad \left(\text{since the lower fn is } y = 0\right)$$

$$= \frac{x^3}{3} + 3x \Big|_{1}^{3}$$

$$= \left(\frac{27}{3} + 9\right) - \left(\frac{1}{3} + 3\right)$$

$$= 18 - \frac{10}{3} = \frac{44}{3}$$

Exercise

Find the area enclosed between the curves $y = x^2 + 2$ and the line $y = 4x - 1$

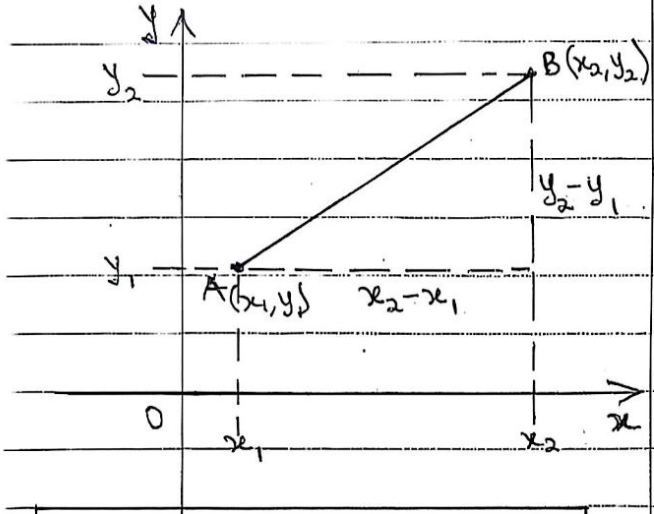
Solution:

Part 4 – Analytic Plane Geometry

- Distance Between 2 points.
- Gradient / Slope of a Line.
- Parallel and Perpendicular Lines
- Equation of straight Lines.

Distance Formula

Given any two pts A and B on a Cartesian plane.



By Pythagoras Rule;

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

or

$$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Examples

① Find the distance between the pts: (a) $(3, -1)$ & $(-2, 5)$ (b) $(-1, 3)$ & $(-8, -4)$ (c) $(3, -4)$ & $(5, 7)$.

Soln:

(a) $A = (3, -1)$ and $B = (-2, 5)$

$\Rightarrow$

$$AB = \sqrt{(3+2)^2 + (-1-5)^2}$$

$$= \sqrt{25 + 36}$$

$$= \sqrt{61} \text{ units}$$

(b) $P = (-1, 3)$ and $Q = (-8, -4)$

$$\Rightarrow PQ = \sqrt{(-1+8)^2 + (3+4)^2}$$

$$= \sqrt{49 + 49} = \sqrt{98} \text{ units}$$

(c) Let $M = (3, -4)$ and $N = (5, 7)$

$\Rightarrow$

$$MN = \sqrt{(3-5)^2 + (-4-7)^2}$$

$$= \sqrt{4 + 121}$$

$$= \sqrt{125} = 5\sqrt{5} \text{ units}$$

② Find the coordinates of a point that is 5 units away from the origin along the x-axis.

Soln:

Along the x-axis, the y-coordinate is 0. $\Rightarrow y_1 = y_2 = 0$

and since the first pt (A) is the origin $\Rightarrow x_1 = 0$

Hence;

$$AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)}$$

$$5 = \sqrt{(0 - x_2)^2 + (0 - 0)^2}$$

$$5 = \sqrt{|x_2|^2}$$

$$\Rightarrow |x_2| = 5 \Rightarrow x_2 = 5 \text{ or } -5$$

∴ the pt will be $(-5, 0)$ or $(5, 0)$

③ If the ~~pts~~ distance between the points $(x, 3)$ and $(7, 6)$ is 5 units, find the possible values of x.

Soln:

Let $A = (x, 3)$ and $B = (7, 6)$ where $AB = 5$.

$$\rightarrow 5 = \sqrt{(x-7)^2 + (3-6)^2}$$

$$2.5 = (x-7)^2 + 9$$

$$\therefore 16 = (x-7)^2$$

$$\Rightarrow x - 7 = \pm\sqrt{16} = \pm 4$$

$$\Rightarrow x = 7 \pm 4$$

$$= 11 \text{ or } 3$$

④ The vertices of a triangle are A(1,2), B(4,6), and C(6,2). Find the lengths of all 3 sides.

Soln:

(i) $AB = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16}$

$= 5$ units

(ii) $BC = \sqrt{(6-4)^2 + (2-6)^2} = \sqrt{4+16}$

$= \sqrt{20}$ units.

(iii) $AC = \sqrt{(6-1)^2 + (2-2)^2} = \sqrt{25+0}$

$= 5$ units

$\Rightarrow \Delta ABC$ is an Isoscales Triangle.

⑤ Find the value of k if the distance between (2, k) and (5, 1) is equal to $\sqrt{13}$.

Soln:

Soln:

$$\sqrt{(5-2)^2 + (1-k)^2} = \sqrt{13}$$

$$\sqrt{9 + (1-k)^2} = \sqrt{13}$$

$$9 + (1-k)^2 = 13$$

$$\Rightarrow (1-k)^2 = 4$$

$$\Rightarrow 1 - k = \pm\sqrt{4} = \pm 2$$

$$\Rightarrow k = 1 \pm 2$$

$\therefore k = -1$ or 3

EXERCISES

① Find the distance between each of the following:

(a) (1,1) and (4,5)

(b) (-3,-4) and (0,0)

(c) origin and (5,-1).

② Prove that the triangle with vertices A(0,0), B(3,4) and C(3,0) is a right-angled triangle.

③ Point A is at (-3,2) and B at (x,5). If the distance AB = 5, find x.

④ Given that points A(x,1) and B(3,5) are 5 units apart, find the possible value(s) of x.

PARALLEL & PERPENDICULAR LINES

(a) Two lines with slopes m_1 and m_2 are parallel to each other if $m_1 = m_2$.

(b) Two lines with slopes m_1 and m_2 are perpendicular of $m_1 = -\frac{1}{m_2}$.

Examples

(1) Find the slope of a line that is perpendicular to the line $3x - 4y = 12$.

Soln:

$3x - 4y = 12$

$\Rightarrow -4y = -3x + 12$

$y = \frac{3}{4}x - 3$

$\Rightarrow$ slope of $3x - 4y = 12$ is $3/4$

Hence, the slope of the $\perp$ar line $= -\frac{1}{3/4} = -\frac{4}{3}$

(2) What is the slope of a line perpendicular to a line of slope $-5/2$?

Soln:

$m = \frac{+1}{+5/2} = \frac{2}{5}$

(3) What is the relationship btwn the lines $y = 2x + 5$ and $4x - 2y = 6$?

Soln:

The slope of $y = 2x + 5$ is 2.

$4x - 2y = 6$

$\Rightarrow 2y = 4x - 6$

$y = 2x - 3$

$\Rightarrow$ slope $= 2$

$\therefore y = 2x + 5$ and $4x - 2y = 6$ are parallel lines, since their slopes are the same.

(4) What is the slope of a line perpendicular to a horizontal line?

Soln:

Horizontal line's slope $= 0$

The line $\perp$ar to horizontal line is the vertical line and the slope of the vertical line $= \frac{y_2 - y_1}{0} =$ Undefined

Exercises

(1) What values of k will make the lines $2x - 3y + 4 = 0$ and $4x + ky - 6 = 0$ parallel?

(A) -6 (B) -3 (C) -4 (D) -2

(2) Find the angle between the lines $y = 2x + 1$ and $3y = -x + 15$

(A) $\tan^{-1}(7/5)$ (B) $\tan^{-1}(11/6)$

(C) $\tan^{-1}(5/7)$ (D) $\tan^{-1}(11/7)$.

(3) Find the $\perp$ar distance between the lines $3x - 4y + 5 = 0$ and $3x - 4y - 7 = 0$

(A) $\frac{12}{5}$ (B) $\frac{5}{12}$ (C) $\frac{7}{5}$ (D) $\frac{4}{5}$

MIDPOINT OF LINES.

Let $M(x_1, y_1)$ and $N(x_2, y_2)$ be the endpoints of a line. The midpoint of line MN, say $P(x, y)$ is given by:

$$P(x, y) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right).$$

Examples

(1) Find the coordinates of the midpoint of the line joining the points $A(-1, 1)$ and $B(7, 9)$

Soln:

$$P(x, y) = \left(\frac{-1+7}{2}, \frac{1+9}{2}\right)$$

$$= (3, 5)$$

(2) Find the point on x-axis that is equidistant from $A(3, -2)$ and $B(5, 4)$.

Soln:

$$P(x, y) = \left(\frac{3+5}{2}, \frac{-2+4}{2}\right)$$

$$= (4, 1)$$

∴ the point equidistant from A and B is $P(4, 1)$ being the midpoint of $\overline{AB}$.

(3) The point $A(x, 1)$ and $B(-6, -5)$ are equidistant from $C(3, -2)$. Find the value(s) of x.

Soln:

$$C(3, -2) = \left(\frac{x-6}{2}, \frac{1-5}{2}\right)$$

$$\Rightarrow \frac{x-6}{2} = 3$$

$$\Rightarrow x = 12$$

EXTERNAL & INTERNAL DIVISION OF LINES INTO SOME RATIOS

This could be done in two ways possible:

(a) Internally: Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be endpoints of a line to be divided into the ratio $m:n$ at point $P(x, y)$. We obtain the coordinates of P using:

$$P(x, y) = \left(\frac{nx_1+mx_2}{n+m}, \frac{ny_1+my_2}{n+m}\right)$$

(b) Externally: If P_1P_2 is divided externally in the ratio $m:n$ at pt $P(x, y)$, then:

$$P(x, y) = \left(\frac{nx_1-mx_2}{n-m}, \frac{ny_1-my_2}{n-m}\right)$$

Examples

(1) If $P(x, y)$ is a pt that divides the line segment whose endpts are $(-3, -6)$ and $(-5, 1)$ internally in the ratio $2:3$, then (a) find $P(x, y)$. (b) Find $P(x, y)$ if the division was externally.

Soln:

$x_1 = -3$, $x_2 = -5$, $y_1 = -6$, $y_2 = 1$, $m = 2$ and $n = 3$

(a) $$P(x, y) = \left(\frac{3(-3)+2(-5)}{3+2}, \frac{3(-6)+2(1)}{3+2}\right)$$

$$= \left(\frac{-19}{5}, \frac{-16}{5}\right)$$

⑥ Externally;

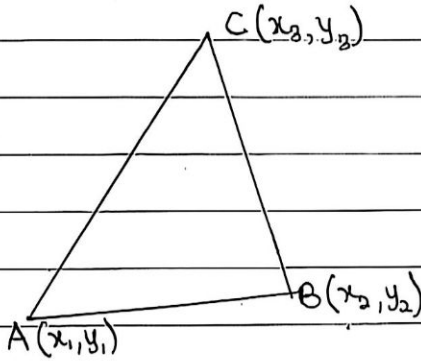
$$P(x,y) = \left(\frac{3(-3)-2(-5)}{3-2}, \frac{3(-6)-2(1)}{3-2}\right)$$

$$= \left(\frac{1}{1}, \frac{-20}{1}\right)$$

$\therefore P(x,y) = (1, -20)$

AREA OF TRIANGLES WITH VERTICES

If the coordinates of the vertices of a triangle are given, say as: $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$



Then the area of ΔABC is given by:

$$= \frac{1}{2}(x_1y_2 - x_2y_1) + \frac{1}{2}(x_2y_3 - x_3y_2) + \frac{1}{2}(x_3y_1 - x_1y_3)$$

$$= \frac{1}{2}(x_1y_2 - x_2y_1 + x_2y_3 - x_3y_2 + x_3y_1 - x_1y_3)$$

N/B: If the pts A, B and C are COLLINEAR (that is, they lie on the same line), the area of ΔABC will be zero.

Examples

① Find the area of a triangle whose vertices are $(-2,1)$, $(4,1)$ and $(7,3)$.

Soln:

$$\text{Area} = \frac{1}{2}[-2(1) - 4(1) + 4(3) - 7(1) + 7(1) - (-2)(3)]$$

$$= \frac{1}{2}[-2 - 4 + 12 - 7 + 7 + 6]$$

$$= \frac{1}{2}(12) = 6 \text{ square units}$$

② Show that the points $A(2,-1)$, $B(0,1)$, $C(-2,3)$ are collinear.

Soln:

Area of $\Delta ABC = 0$ if collinear pts.

where

$$\text{Area} = \frac{1}{2}[2(1) - 0(-1) + 0(3) - (-2)(1) + (-2)(-1) - 2(3)]$$

$$= \frac{1}{2}[2 - 0 + 0 + 2 + 2 - 6]$$

$$= 0$$

$\therefore$ A, B and C are collinear.

EXERCISE

① Find the area of ΔPQR if $P = (2,-7)$, $Q = (7,3)$ and $R = (9,7)$.

② Determine whether or not the vertices of triangle ABC given by $A(2,-1)$, $B(5,3)$ and $C(-4,-9)$ ~~are~~ lie on the same straight line.

EQUATIONS OF STRAIGHT LINES.

SLOPE OF A LINE (m)

Slope, $m = \frac{\text{rise}}{\text{run}}$

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

Examples

① Find the slope of the line that passes through the points A(4, -3) and B(7, 1).

Soln:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - (-3)}{7 - 4}$$

$$= \frac{4}{3}$$

② Slope of a horizontal line.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

since its horizontal, $y_1 = y_2$ (no rise)

$$\Rightarrow m = \frac{0}{x_2 - x_1} = 0$$

③ Slope of a vertical line.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

since it is vertical (no run)

$$\Rightarrow x_2 = x_1$$

$$\Rightarrow m = \frac{y_2 - y_1}{0} = \text{undefined} = \infty$$

④ Find the slope of a line that cuts the x-axis at $x = 3$ and passes through the point (6, 6).

Soln:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 0}{6 - 3}$$

where $y = 0$ on the x-axis.

$$\Rightarrow m = 2$$

⑤ Find the slope of a line that has x and y intercepts as 2 and -1 respectively.

Soln:

the pts are (0, -1) and (2, 0)

at x-intercept, $y = 0$

and $x = 0$ at y-intercept.

Hence,

$$\text{Slope, } m = \frac{0 - (-1)}{2 - 0}$$

$$= 1/2$$

⑥ Find the slope of a line that passes through the origin and point A(-2, 5).

Soln:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

where $x_1 = 0$, $y_1 = 0$ (origin)

$x_2 = -2$, $y_2 = 5$

$$\Rightarrow m = \frac{5 - 0}{-2 - 0} = 5/-2$$

EQUATION OF LINES.

The eqn of a straight line can be written in several forms, depending on the available information. We will consider about 7 forms or cases of straight line equations, along with the required information.

(1) SLOPE-INTERCEPT FORM.

Eqn: $y = mx + c$

where m = slope

c = y-intercept.

*We use this when we are given m and c.

(2) Point-Slope Form

Equation: $y - y_1 = m(x - x_1)$

where

m = slope

(x_1, y_1) = given point.

N/B: We use this eqn form when the slope and a point are given.

(3) Two-Point Form

Eqn:

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$$

where (x_1, y_1) and (x_2, y_2) are the 2 given points.

N/B: This is used when two pts are given. It is similar to the second case.

(4) GENERAL Form (Standard Form)

Eqn: $Ax + By + C = 0$

where

A, B and C are real nos.

A & B must not both be zeros.

(5) Intercept Form

Eqn: $\frac{x}{a} + \frac{y}{b} = 1$

where:

a = x-intercept

b = y-intercept.

N/B: We use this eqn form when the x and y intercepts only are given.

(6) Vertical Line Form

Eqn: $x = a$

⇒ a vertical line passing through $x = a$ or pt $(a, 0)$.

where

m = undefined.

(7) Horizontal Line Form.

Eqn: $y = b$.

⇒ a horizontal line passing through $y = b$

where slope, $m = 0$.

Assignment

Summarize in a table of 3 columns the 7 forms or cases of straight line eqns showing:

- The Form
- The equation
- Given or Needed Information.

Worked Examples

① Find the equation of a line passing through $(1, 2)$ with slope 3.

Soln:

A point-slope case, using

$y - y_1 = m(x - x_1)$

$\Rightarrow y - 2 = 3(x - 1)$

$\Rightarrow y = 3x - 1$

or $y - 3x + 1 = 0$

② Find the eqn if the line passes through $(3, -4)$ with slope 2.

Soln:

Using $y - y_1 = m(x - x_1)$

where $x_1 = 3$ and $y_1 = -4$

$\Rightarrow y + 4 = 2(x - 3)$

$y = 2x - 6 - 4$

$\therefore y = 2x - 10$

③ Find the eqn of line passing through the pts $(1, 2)$ and $(3, 6)$.

Soln:

Form: Two-point

using $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$\Rightarrow y - 2 = \frac{6 - 2}{3 - 1}(x - 1)$

$\Rightarrow y - 2 = 2(x - 1)$

$\Rightarrow y = 2x - 2 + 2$

$\therefore y = 2x$ or $y - 2x = 0$

④ A line has x-intercept 4 and y-intercept -2, find its eqn.

Soln:

Form: Intercept Form

using $\frac{x}{a} + \frac{y}{b} = 1$

where $a = 4$ and $b = -2$

$\Rightarrow \frac{x}{4} - \frac{y}{2} = 1$

we can modify to general form; multiply through by 8.

$2x - 4y = 8$

$\Rightarrow 2x - 4y - 8 = 0$

⑤ A horizontal line passes through the point $(5, -3)$, what is its eqn?

Soln:

using $x = a$ where $a = 5$

$\therefore$ eqn is $x = 5$

⑥ Find the eqn of the line perpendicular to $y = 3x - 1$ and passing through $(6, 5)$.

Soln:

A line passing through $(6, 5)$ has

$x_1 = 6$ and $y_1 = 5$

since its $\perp$ar to $y = 3x - 1$

$\Rightarrow$ its slope, $m = -\frac{1}{3}$

Hence,

Eqn: $y - y_1 = m(x - x_1)$

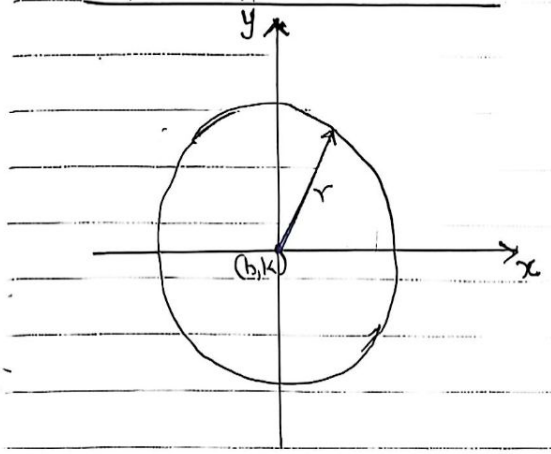
$y - 5 = -\frac{1}{3}(x - 6)$

$3y - 15 = -1(x - 6)$

$3y - 15 = -x + 6$

$\therefore$ The eqn is: $3y + x - 21 = 0$

EQUATION OF CIRCLES.



Let the centre = (h,k) and radius = r

The standard eqn of a circle is given as:

$$\boxed{(x-h)^2+(y-k)^2 = r^2}$$

NB (i) If the centre of the circle is at the origin, then eqn is:

$$x^2+y^2 = r^2$$

i.e $O = (0,0) = (h,k)$.

(ii) The general form which is obtained from expanding the brackets in the standard form, is given as:

$$\boxed{x^2+y^2+Dx+Ey+F=0}$$

Examples

(1) Find the radius and center of $(x-3)^2+(y+2)^2 = 16$

Soln:

h = the value of x when $x-3=0$

$\Rightarrow h = 3$

k = the value of y when $y+2=0$

$\Rightarrow k = -2$

∴ Centre $= (3,-2)$

and radius $= \sqrt{16} = 4$

(2) Find the radius and center of $(x-\frac{1}{2})^2+(y-1)^2 = 12$

Soln:

$h = \frac{1}{2}$ and $k = 1$ $\Rightarrow$ Centre $= (\frac{1}{2}, 1)$

radius $= \sqrt{12} = 2\sqrt{3}$

(3) Find the standard form of the eqn of the circle with center at $(2,-5)$ and radius 6.

Soln:

$h = 2$, $k = -5$ and $r = 6$

Eqn:

$(x-2)^2+(y+5)^2 = 6^2$

$\Rightarrow (x-2)^2+(y+5)^2 = 36$

(4) Find the eqn of a circle centred at the origin with radius 7.

Soln:

$r = 7$ and center $= (0,0)$

$\Rightarrow$ Eqn: $x^2+y^2 = 7^2$

$x^2+y^2 = 49$

(5) What is the general form of the circle $(x-1)^2+(y-2)^2 = 9$?

Soln:

$(x-1)^2+(y-2)^2-9 = 0$

$x^2-x-x+1+y^2-2y-2y+4-9 = 0$

$x^2+y^2-2x-4y+1+4-9 = 0$

∴ Eqn: $x^2+y^2-2x-4y-4 = 0$

RADIUS AND CENTER FROM GENERAL EQUATIONS

If the eqn of a circle is in the general form, we determine the radius and center using methods:

(A) Completing Square Method.

Example:

Find the center and radius of $x^2 + y^2 + 4x - 6y + 9 = 0$

Soln:

$x^2 + 4x + y^2 - 6y = -9$

add to both the square of the half of the coefficients of x & y.

$x^2 + 4x + 2^2 + y^2 - 6y + (-3)^2 = -9 + 4 + 9$

gather the squares on LHS

$(x+2)^2 + (y-3)^2 = 4$

and that gives the standard form.

$\therefore$ Center $= (-2, 3)$ and

radius $= \sqrt{4} = 2$

(B) Easy Approach:

From $x^2 + y^2 + Dx + Ey + F = 0$

$h = -\frac{D}{2}$ and $k = -\frac{E}{2}$

$r = \sqrt{h^2 + k^2 - F}$

Example: Repeat the example in (A) above.

Soln:

$x^2 + y^2 + 4x - 6y + 9 = 0$

$\Rightarrow D = 4, \quad E = -6, \quad F = 9$

Hence;

$h = -\frac{4}{2} = -2; \quad k = -\frac{(-6)}{2} = 3$

$\therefore$ Center $= (-2, 3)$

and;

Radius, $r = \sqrt{(-2)^2 + 3^2 - 9}$

$= \sqrt{4 + 9 - 9} = 2$

Ex: What is the radius of the circle $x^2 + y^2 + 6x + 8y + 9 = 0$?

Soln:

$r = \sqrt{h^2 + k^2 - F}$

where $h = -\frac{6}{2} = -3$

$k = -\frac{8}{2} = -4$ & $F = 9$

$\Rightarrow r = \sqrt{9 + 16 - 9} = 4$

EXERCISE

① Write the standard and general form of the eqn of a circle with center $(2, -5)$ and radius 6.

Ans: (i)

(ii)

② What is the radius of the circle $x^2 + y^2 + 6x + 8y + 9 = 0$?

(A) 2 (B) 3 (C) 4 (D) 5

③ The center of $x^2 + y^2 - 10x + 6y = 7$ is:

(A) $(5, -3)$ (B) $(-5, 3)$ (C) $(-5, -3)$ (D) $(5, 3)$.

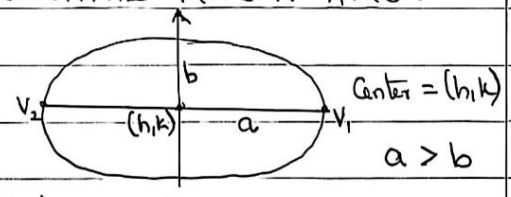
④ Find dy/dx from a circle whose center is $(3, 0)$ and radius 5.

Ans:

EQUATION OF ELLIPSE

(A) ~~Horizon~~ There are two standard forms of the equation of ellipse, which depends on the orientation of the major axis:

(A) HORIZONTAL MAJOR AXIS:



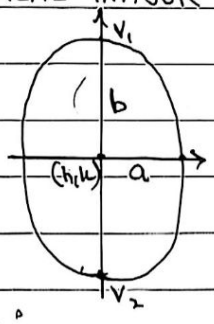
$a > b$

Equation:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$c = \sqrt{a^2 - b^2}$, vertices $= (\pm a, 0)$

(B) VERTICAL MAJOR AXIS:



Center $= (h,k)$

$a > b$

Equation:

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

$c = \sqrt{a^2 - b^2}$, Vertices $= (0, \pm a)$.

and generally, the Foci are given by:

$F = (h \pm c, k)$ – Horizontal.

or

$F = (h, k \pm c)$ – Vertical.

NB: If the ellipse is at the origin, then foci will be

$$F = (\pm c, 0) \text{ or } (0, \pm c)$$

Examples

(1) Find the coordinates of the vertices and foci of $\frac{x^2}{100} + \frac{y^2}{64} = 1$

Soln:

$\frac{x^2}{100} + \frac{y^2}{64} = 1 \Rightarrow \frac{x^2}{10^2} + \frac{y^2}{8^2} = 1$

$\Rightarrow a = 10, \ b = 8, \ h = k = 0.$

(a) Coordinates of Vertices $= (\pm a, 0)$
$= (\pm 10, 0)$

(b) Foci $= (h \pm c, k)$

where $c = \sqrt{a^2 - b^2} = \sqrt{100 - 64}$
$= 6$

$\therefore$ Foci $= (0 \pm 6, 0)$
$= (\pm 6, 0)$

(2) Identify the conic section: $9x^2 + 16y^2 = 144$. Find the vertices and foci.

Soln:

$9x^2 + 16y^2 = 144$

divide through by 144.

$\frac{x^2}{16} + \frac{y^2}{9} = 1$ — An Ellipse

$\Rightarrow a = 4, \ b = 3, \ h = 0, \ k = 0$

(a) Vertices $= (\pm a, 0)$
$= (\pm 4, 0)$

(b) Foci $= (h \pm c, k)$

where $c = \sqrt{16 - 9} = \sqrt{7}$

$= (\pm \sqrt{7}, 0)$

(3) What is the eqn of an ellipse with center at $(2,-1)$, $a=4$, $b=3$, and horizontal major.

Soln:

using $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

where $h=2$ and $k=-1$

we have;

$$\frac{(x-2)^2}{4^2} + \frac{(y+1)^2}{3^2} = 1$$

$$\Rightarrow \frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$$

OR

$$9(x-2)^2 + 16(y+1)^2 = 144$$

(4) Find the eqn of an ellipse with minor axis of length 8 and a vertex at $(0,-5)$.

Soln:

Length of minor axis $= 8 = 2b$

$\Rightarrow b = 4$

A vertex is at $(0,-5) \Rightarrow a = 5$

Note, since the vertex is on the y-axis, the major axis is vertical and assuming the center = origin

$\Rightarrow$ using $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

we have; $\frac{x^2}{16} + \frac{y^2}{25} = 1$

OR

$$25x^2 + 16y^2 = 400$$

(6) Foci of the above ellipse $= (h, k \pm c)$

where $c = \sqrt{a^2 - b^2} = \sqrt{9} = 3$

$\therefore$ Foci $= (0, \pm 3)$

(5) Let $25(x-3)^2 + (y+2)^2 = 25$ be an ellipse. Find!

(a) the vertices

(b) the Foci

(c) the center of the ellipse.

Soln:

From the eqn; $\frac{(x-3)^2}{1} + \frac{(y+2)^2}{25} = 1$

$h=3$, $k=-2$

$a=5$ and $b=1$

(a) Vertices $= (0, \pm a)$

$= (0, \pm 5)$

(b) Foci $= (h, k \pm c)$

where $c = \sqrt{a^2 - b^2} = \sqrt{24}$

$= (3, -2 \pm \sqrt{24})$

$= (3, -2+\sqrt{24})$ or $(3, -2-\sqrt{24})$